Sampling and Variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Sampling and Variability

> Prepare Sampling and Variability: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Sample from a distribution using explicit **cumulative interval**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Tokens and Context

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

Look for: Account for fixed text, **evidence** and reserved **output** under a toy budget

Reference example, not the only acceptable wording: Teacher explanation: 100-20-10-20 = 50 supplied token units.

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Reference example, not the only acceptable wording: Teacher explanation: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

Concept to explain

Decoding choices trade consistency, diversity, and risk.

Term	Meaning
decoding	Selecting output tokens from a model's scores using a specified procedure.
variability	Differences observed across repeated runs, samples or conditions.
temperature intuition	A rough understanding of how temperature rescales token scores: higher values commonly spread choices more, without guaranteeing creativity or truth.
seeded test	A test containing a deliberately planted, known condition or error to check whether it is handled.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

No. Later conditional probabilities can change path rankings, as the two-step counterexample shows.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

Apply the connected idea: A small attention calculation

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

Normalise the permitted weights by their remaining total; handle an empty set explicitly.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can calculate a toy **temperature** change and trace a **sampling** draw.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Sampling and Variability using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Low **variability** favors high-weight continuations repeatedly.

2. Higher **variability** gives lower-weight options more chances.

3. Diversity may help brainstorming but can increase off-task or unsupported detail.

4. Choose settings by task, then evaluate repeated runs—not one favorite **output**.

Recalculate, retrace, or restate the reasoning in your own representation:

> CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> ### The teaching case
>
> One prompt produces three different valid quiz questions.

> Does **variability** mean failure?

Exact explanation to model

> ### Why the result follows
>
> No. **Sampling** can vary **output**s. **Score** correctness, relevance and format over repeated trials; consistency differs from quality.

Explain the added worked example

Explain the added worked example · 1

Start with probabilities red 0.6, blue 0.3, green 0.1 in that fixed order. Greedy chooses red. **Sampling** uses a draw u with 0 <= u < 1 and **cumulative interval**s: red [0,0.6), blue [0.6,0.9), green [0.9,1).

Left endpoints are included, right endpoints excluded. Thus u=0.6 selects blue and u=0.9 selects green. The draw 1 is outside this contract.

Explain the added worked example · 2

Sampling does not mean choosing all tokens equally. Across many independent draws the frequencies may approach the specified probabilities, but a small sample need not match them exactly.

Fixed recorded draws allow a reproducible classroom trace. A random seed and generator version help replay a real **sampling** program, not establish the truth of **output**s.

Explain the added worked example · 3

For strictly positive probabilities p_i and **temperature** T>0, define q_i = p_i^(1/T) / sum_j p_j^(1/T). This classroom formula is equivalent to scaling logits before a **softmax** when p came from those logits.

We use it directly on the supplied table. At T=1 the distribution is unchanged. At T=0.5, square each probability: 0.36,0.09,0.01; total 0.46. The adjusted distribution is red 18/23, blue 9/46, green 1/46.

Key terms

Term	Meaning
Sampling	Selecting an alternative using a specified random procedure
Greedy choice	Selecting an alternative with largest next-token probability under a tie rule
Temperature	A positive parameter that changes the concentration of a distribution under a stated transformation
Cumulative interval	A range of draws assigned to an alternative using accumulated probability mass

Apply the connected idea: A small attention calculation

Apply the connected idea: A small attention calculation · 4

For one query q and keys k_i of width d, compute **score** s_i = dot(q,k_i)/sqrt(d). Then obtain weights from **softmax** and combine value vectors. Dot products may be negative; probabilities cannot.

The **softmax** performs the conversion. This is a single-head classroom calculation, not a full language model.

Apply the connected idea: A small attention calculation · 5

Define **softmax** weights as exp(s_i-m)/sum_j exp(s_j-m), where m is the maximum permitted **score**.

Subtracting the same maximum from every permitted **score** leaves the ratios unchanged and avoids unnecessary overflow from large positive **score**s.

Natural exponent exp is introduced here; use a calculator or Python math.exp rather than estimating its meaning from the word exponential.

Apply the connected idea: A small attention calculation · 6

Take q=[1] and d=1. The scale sqrt(1) is 1, so **score**s are 0 and ln(3). ln(3) is the number whose natural exponential is 3, approximately 1.098612. Exponentials have ratio 1:3, hence weights 1/4 and 3/4.

Output is (1/4)*2 + (3/4)*6 = 5. Using maximum subtraction gives exponentials 1/3 and 1 and the same normalised weights.

Guided: shifted **score**s -ln(3),0; exponentials 1/3,1; weights .25,.75; **output** .25*4+.75*8 = 7. Exit: both coefficients are .5; opposite values cancel to zero.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 7

The lower positive **temperature** concentrates this table on its largest entry. For T=2, use square roots before normalising; this flattens this unequal distribution.

These changes leave the order of positive probabilities unchanged, so greedy still selects red. They change the **sampling** distribution, not the factual **evidence** behind a generated claim.

Model the reasoning aloud · 8

Our formula excludes T=0; dividing by zero is undefined. Some systems offer a zero-**temperature** setting as a separate greedy policy, but that is not a value to substitute into this formula.

We also avoid claiming that any one setting removes all nondeterminism in an unspecified system.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sampling and Variability

A fictional two-token distribution illustrates **variability** in generation.

The facts we will use

> **Supplied facts**
>
> At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

> **Supplied facts**
>
> Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

AI ACADEMY / AI-3/4

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.058823529411764705

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/4

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Original distribution

A boundary 0.8; u=0.85 selects B.



2. T=0.5 weights

Square probabilities:0.64 and 0.04; total 0.68.



3. Normalise

A=16/17≈0.9412; B=1/17≈0.0588.



4. Same draw

0.85 is below 16/17, so T=0.5 selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Original distribution	A boundary 0.8; u=0.85 selects B.
2. T=0.5 weights	Square probabilities:0.64 and 0.04; total 0.68.
3. Normalise	A=16/17≈0.9412; B=1/17≈0.0588.
4. Same draw	0.85 is below 16/17, so T=0.5 selects A.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

What this example does not establish

Keep the stated limit

Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

Change one thing in the pictured case

Change one thing

Change: Set the draw to 0.97, keeping both distributions.

Predicted result

Prediction: Both **temperature**s select B.

Why the result follows

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Ask while showing the picture

Ask while showing the picture · 9

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Sampling and Variability, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: One prompt produces three different valid quiz questions.

2. State the decision to explain

Write the question to explain: Does **variability** mean failure?

3. Reveal the worked reasoning

Reveal after **prediction**s: No. **Sampling** can vary **output**s. **Score** correctness, relevance and format over repeated trials; consistency differs from quality.

Hold back the independent task

Keep for an independent attempt: Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: T=1 gives red,blue,green; T=0.5 gives red,red,blue. Three fixed draws check the procedure, not expected long-run proportions.

Exit: concentration changed; the claim of impossible factual errors lacks source-grounded evaluation and cannot be established by **temperature** alone.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read section 1 and complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the worked case in section 2 and complete W2.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.

Stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read section 1 and complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the worked case in section 2 and complete W2.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion practice

Completion task: At T=0.5 compute squared weights, total and normalised probabilities. Check the sum.

Reasoning to listen for

Teacher check: Teacher explanation: Weights 0.36,0.09,0.01 sum 0.46. Probabilities 18/23,9/46,1/46 sum 1.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Sampling** and **temperature**. Proposed explanation: "**Sampling** means equal chances". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Intervals have widths equal to the probabilities; unequal widths give unequal chances.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: A small attention calculation

Target: Compute scaled query-key **score**s and **softmax** weights

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Use values [2],[6] for the preceding weights. Compute **output** and **output** when the second key is excluded.

> #### Reasoning to listen for
>
> Teacher check: Teacher explanation: Full **output** 5; with B excluded, weight A=1 and **output** 2.

Individual check: For A small attention calculation, use its worked case to evaluate this proposed explanation: “**Score**s already are probabilities”. Explain the deciding step.

> **Expected repair**
>
> Expected correction: **Score**s may be negative or fail to sum to one; **softmax** supplies normalised nonnegative weights.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

> Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

Reasoned teacher answer

No. B still has probability 1/17, and the table supplies no truth labels for either token.

Changed-case reasoning: Both **temperature**s select B. 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Sampling and Variability investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Run each setting five times on one fixed prompt, **score** consistency, usefulness, and unsupported detail, then justify a task-specific choice.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Sampling means equal chances: Intervals have widths equal to the probabilities; unequal widths give unequal chances.

T=0 can be inserted into p^(1/T): The formula requires T>0; greedy is a separate policy.

Lower **temperature** verifies claims: A probability transformation does not add factual **evidence**.

Check these misconceptions

Scores already are probabilities: **Score**s may be negative or fail to sum to one; **softmax** supplies normalised nonnegative weights.

Value width must equal key width: Values share a common **output** width, which can differ from query/key width.

Zero **output** means no source contributed: Nonzero weighted contributions may cancel.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

TEACHER GUIDE · 5 OF 9

Assess and explain the answers

> Use the Sampling and Variability questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

No. **Sampling** can vary **output**s. **Score** correctness, relevance and format over repeated trials; consistency differs from quality.

Independent question

Independent question · withhold the key

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

Expected independent answer

Teacher key · Expected independent answer

Sampling can produce variation with an unchanged prompt. Run both prompts on the same tasks, record settings, repeat trials and judge with the same rubric.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Any difference between **output**s proves a difference between prompts.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

A is consistent but 0/3 correct; B is less consistent but 2/3 correct on these trials. Repeat on predefined varied cases and keep conditions fixed. Identical **output** is not **evidence** of quality.

Explanations for the additional workbook examples

W1 Trace boundaries

Workbook question · W1

At T=1 select tokens for u=0,0.6,0.9 and 0.999. Explain the handling of u=1.

Teacher reasoning · W1

Teacher explanation: Selections red,blue,green,green. The draw 1 violates the half-open [0,1) **input** contract and must be rejected.

W2 Compute the adjusted table

Workbook question · W2

At T=0.5 compute squared weights, total and normalised probabilities. Check the sum.

Teacher reasoning · W2

Teacher explanation: Weights 0.36,0.09,0.01 sum 0.46. Probabilities 18/23,9/46,1/46 sum 1.

Teacher reasoning · W2

Under the lesson's supplied **temperature** procedure, T=0.5 means exponent 1/T=2. Square the weights to obtain 0.36, 0.09 and 0.01; their total is 0.46. Divide each by 0.46 to obtain 18/23, 9/46 and 1/46.

Converting 18/23 to 36/46 shows that the sum is (36+9+1)/46 = 1.

W3 Compare a fixed draw

Workbook question · W3

Trace u=0.7 at T=1 and T=0.5. Does the different **output** demonstrate improved factual accuracy?

Teacher reasoning · W3

Teacher explanation: T=1 selects blue; T=0.5 selects red. This follows changed intervals and supplies no factual correctness **evidence**.

W4 Sample a new table independently

Workbook question · W4

Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

Teacher reasoning · W4

Teacher explanation: Squared weights 0.25,0.09,0.04 total 0.38. Probabilities 25/38,9/38,4/38; boundaries 25/38 and 34/38. T=1: 0.5 selects B,0.9 selects C. T=0.5: 0.5 selects A,0.9 selects C because 0.9 > 34/38.

The trace shows distribution effects, not factual accuracy.

W5 Specify a valid experiment

Workbook question · W5

A student changes the prompt and model as well as **temperature**, then attributes a better answer to **temperature**. Propose a fair comparison and reject any unsupported universal claim.

Teacher reasoning · W5

Teacher explanation: Fix prefix/prompt, model/version, candidate distribution and **sampling** policy while varying **temperature**. Record draws and evaluate stated task criteria across cases.

The mixed comparison cannot isolate **temperature**; even a controlled result cannot prove all future **output**s correct.

A small attention calculation — W1 Compute scores and weights

For q=[1], keys [0],[ln(3)] and d=1, compute **score**s and **softmax** weights.

Teacher reasoning

Teacher explanation: **Score**s 0,ln(3); exponential ratio 1:3 gives weights 1/4,3/4.

The dot products are 1*0=0 and 1*ln(3)=ln(3). Division by sqrt(d)=sqrt(1)=1 leaves them unchanged. Exponentiation gives 1 and 3, with total 4. Normalising gives weights 1/4 and 3/4.

These are compatibility weights under the stated calculation, not truth **score**s.

A small attention calculation — W2 Combine and mask

Use values [2],[6] for the preceding weights. Compute **output** and **output** when the second key is excluded.

Teacher reasoning

Teacher explanation: Full **output** 5; with B excluded, weight A=1 and **output** 2.

A small attention calculation — W3 Trace equal scores

For q=[1,1], keys [1,0],[0,1], values [2,0],[0,4], compute scaled **score**s, weights and **output**.

Teacher reasoning

Teacher explanation: Both **score**s 1/sqrt(2); weights .5,.5; **output** [1,2].

The query has dot product 1 with each key. Both **score**s are therefore 1/sqrt(2). Equal **score**s give equal **softmax** weights, 0.5 each. Combine the values coordinate by coordinate: 0.5*[2,0] + 0.5*[0,4] = [1,2].

A small attention calculation — W4 Calculate new attention independently

q=[1], keys [0],[ln(2)],[ln(3)], values [0],[3],[8], width 1. Compute **score**s, weights and **output**. Then mask the third item and recompute. Explain whether **output** establishes a factual claim.

Teacher reasoning

Teacher explanation: **Score**s 0,ln(2),ln(3); weights 1/6,2/6,3/6. **Output** 0+1+4=5. Mask third: weights 1/3,2/3,0; **output** 2. Neither weighted **output** verifies a factual claim.

A small attention calculation — W5 Check stability and shape

Scores are 1000 and 1000+ln(3); values 4 and 8. Compute stable shifted weights and **output**. What if one key has width two and the query width one?

Teacher reasoning

Teacher explanation: Shift to -ln(3),0; weights 1/4,3/4 and **output** 7. Mismatched key/query widths must be rejected rather than silently truncating.

Assess independent transfer

Independent question · withhold the key

Selected case: Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Squared weights 0.25,0.09,0.04 total 0.38. Probabilities 25/38,9/38,4/38; boundaries 25/38 and 34/38. T=1: 0.5 selects B,0.9 selects C. T=0.5: 0.5 selects A,0.9 selects C because 0.9 > 34/38.

The trace shows distribution effects, not factual accuracy.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Sampling and Variability to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Use low **variability** for structured study answers and allow more diversity only for labeled brainstorming.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 5: Prompt as a Contract. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires probabilities 25/38,9/38,4/38, correct boundaries and B/C versus A/C. Fresh retry: A=0.4,B=0.4,C=0.2; at T=0.5 probabilities 4/9,4/9,1/9. Draws 0.4 and 0.85 select B,C at T=1 but A,B at T=0.5.

Greedy ties must use an explicit rule.

Week 5 connects token IDs to embedding vectors and introduces position information without treating IDs as semantic measurements.

Readiness and reassessment for A small attention calculation

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires weights 1/6,1/3,1/2 and **output**s 5 then 2. Fresh retry: keys [0],[ln(3)],[ln(4)], values [2],[6],[0]. Weights 1/8,3/8,4/8; **output** 2.5. Exclude third: weights 1/4,3/4,0; **output** 5.

Plan the delayed return

Suggested return points in this level: ai-3/5, ai-3/7, ai-3/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Sampling and Variability lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Sample from a distribution using explicit **cumulative interval**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: At T=0.5 compute squared weights, total and normalised probabilities. Check the sum.

Supported practice answer

Reasoned practice answer: Teacher explanation: Weights 0.36,0.09,0.01 sum 0.46. Probabilities 18/23,9/46,1/46 sum 1.

Check every learner before discussion: Use the worked case for **Sampling** and **temperature**. Proposed explanation: "**Sampling** means equal chances". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Intervals have widths equal to the probabilities; unequal widths give unequal chances.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Set the draw to 0.97, keeping both distributions. Require a **prediction** and a reason before comparing. Expected result: Both **temperature**s select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

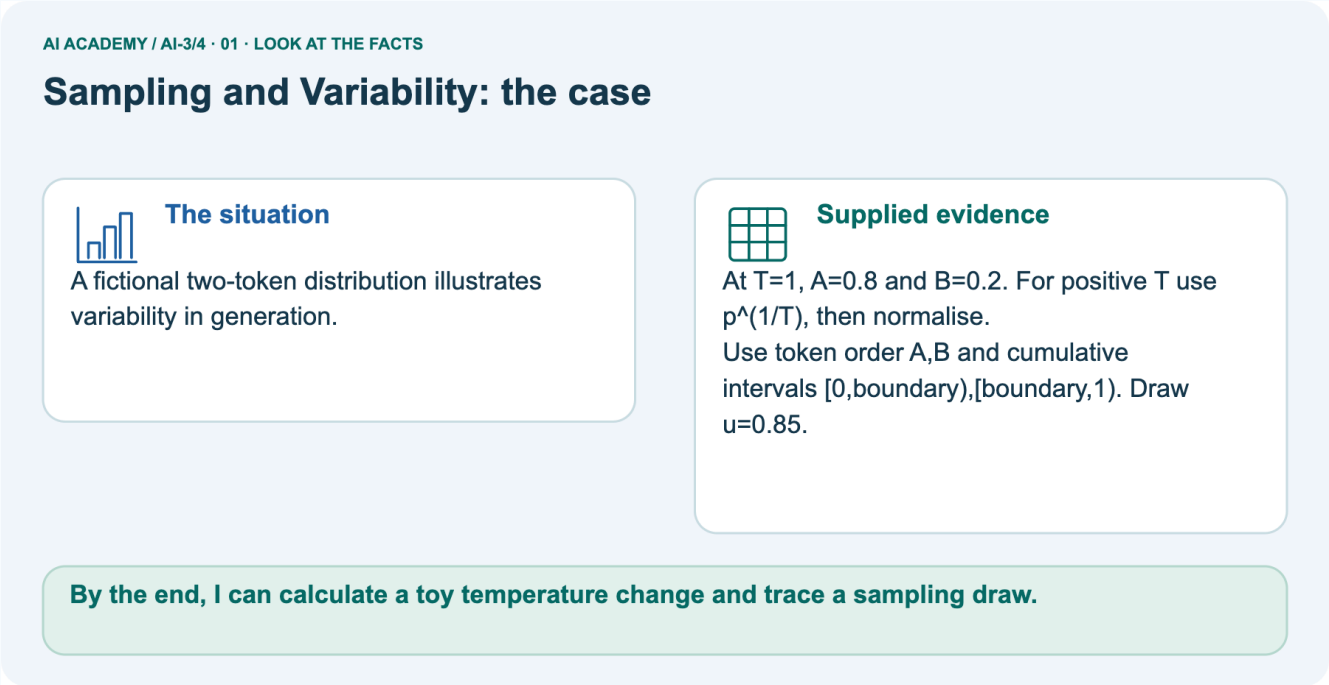
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Sampling and Variability graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sampling and Variability: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional two-token distribution illustrates **variability** in generation. Supplied facts: At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

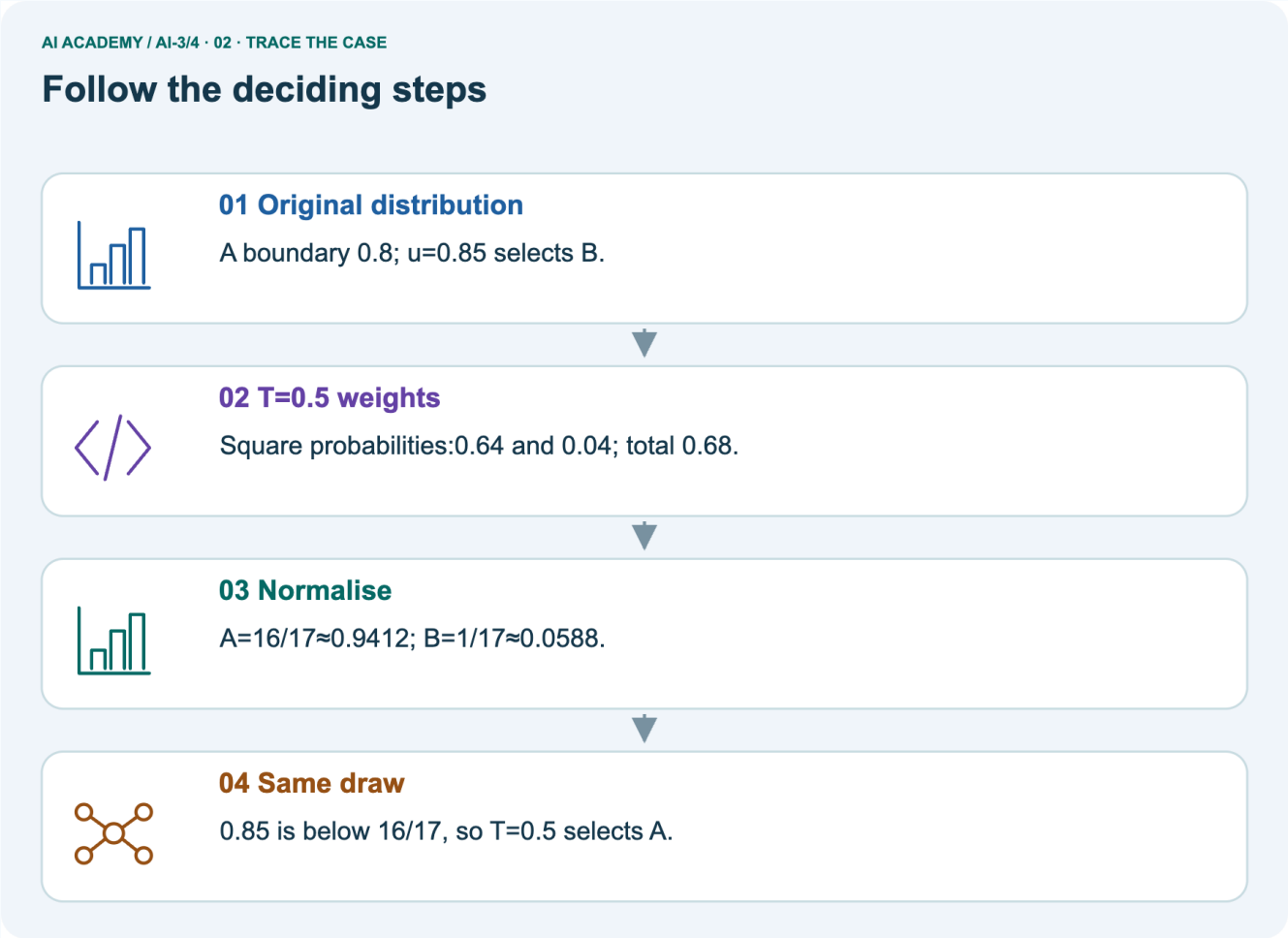
> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

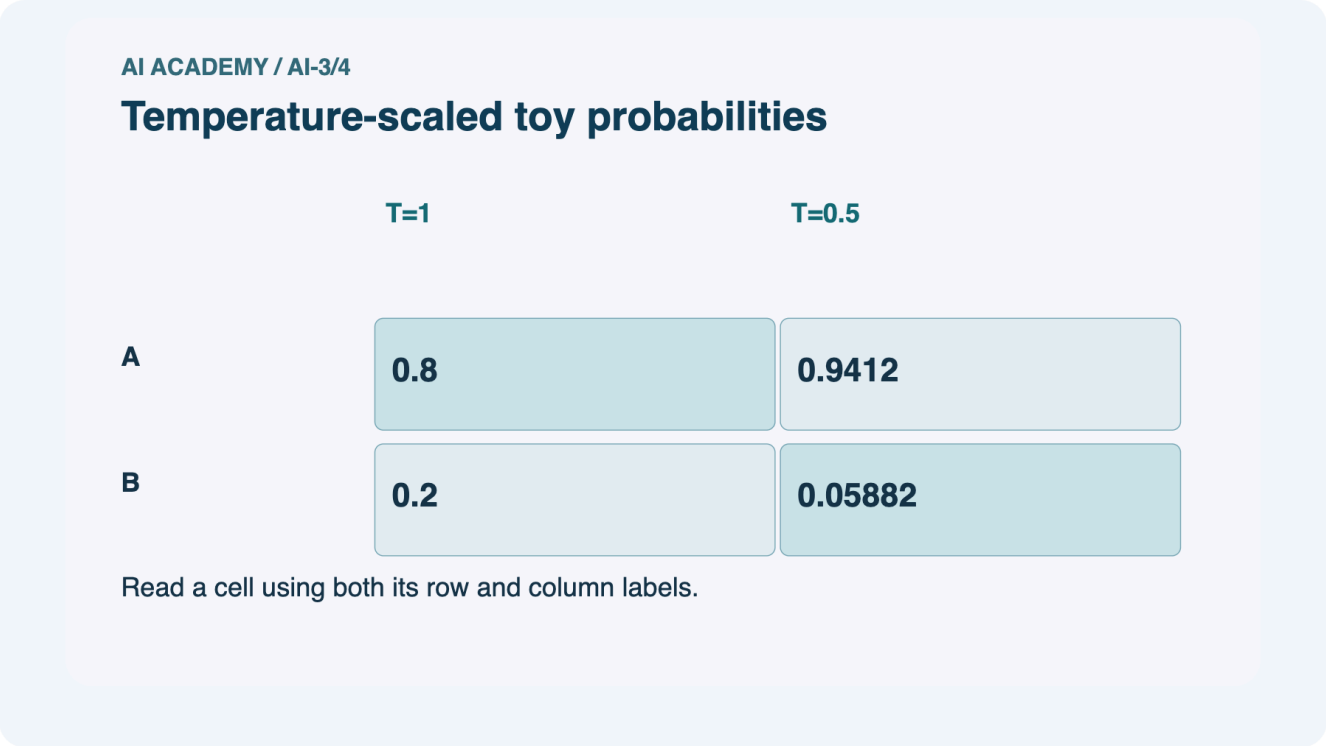
Original distribution: A boundary 0.8; u=0.85 selects B. → T=0.5 weights: Square probabilities:0.64 and 0.04; total 0.68. → Normalise: A=16/17≈0.9412; B=1/17≈0.0588. → Same draw: 0.85 is below 16/17, so T=0.5 selects A.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Temperature-scaled toy probabilities



Follow the labelled picture with the matching explanation. Open the vector diagram

Temperature-scaled toy probabilities. A: T=1 = 0.8, T=0.5 = 0.9411764705882353. B: T=1 = 0.2, T=0.5 = 0.058823529411764705.

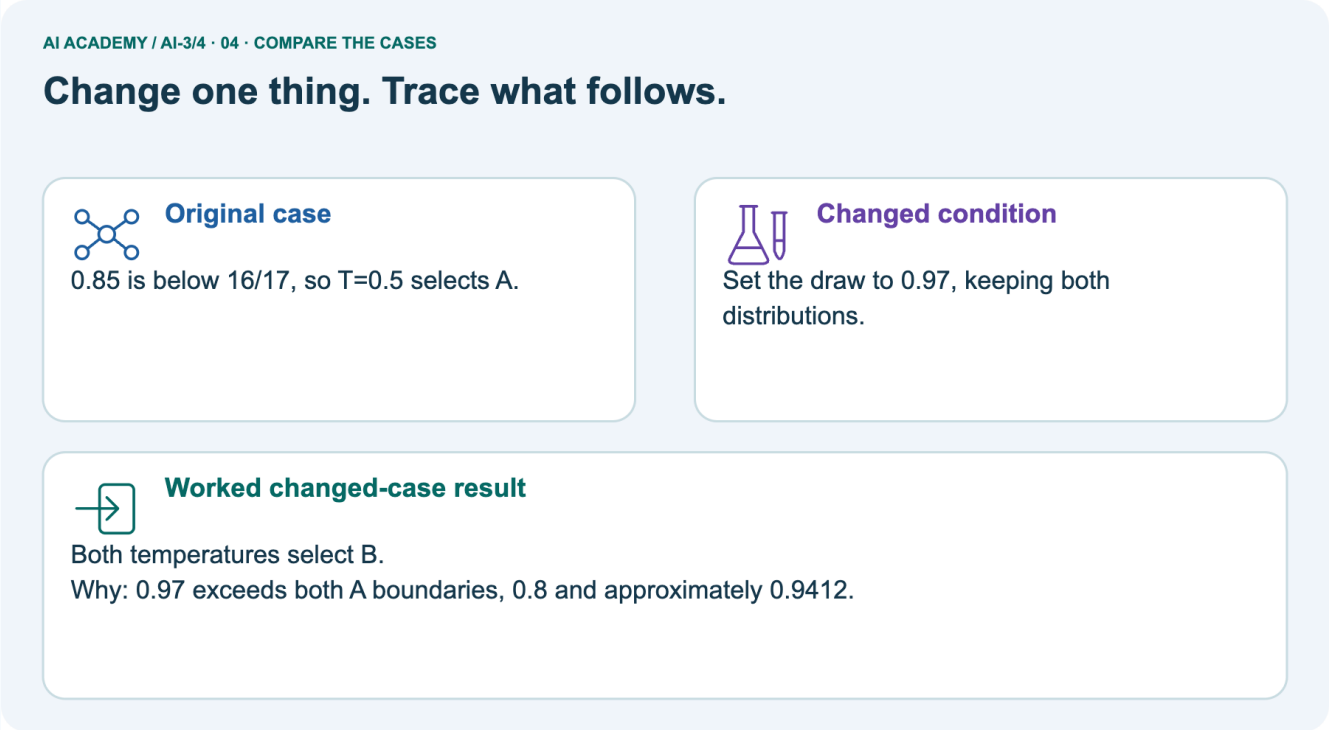
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 0.85 is below 16/17, so T=0.5 selects A. Changed condition: Set the draw to 0.97, keeping both distributions. Expected result: Both **temperature**s select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Predict before checking the worked changed-case result. Name what changed and what the

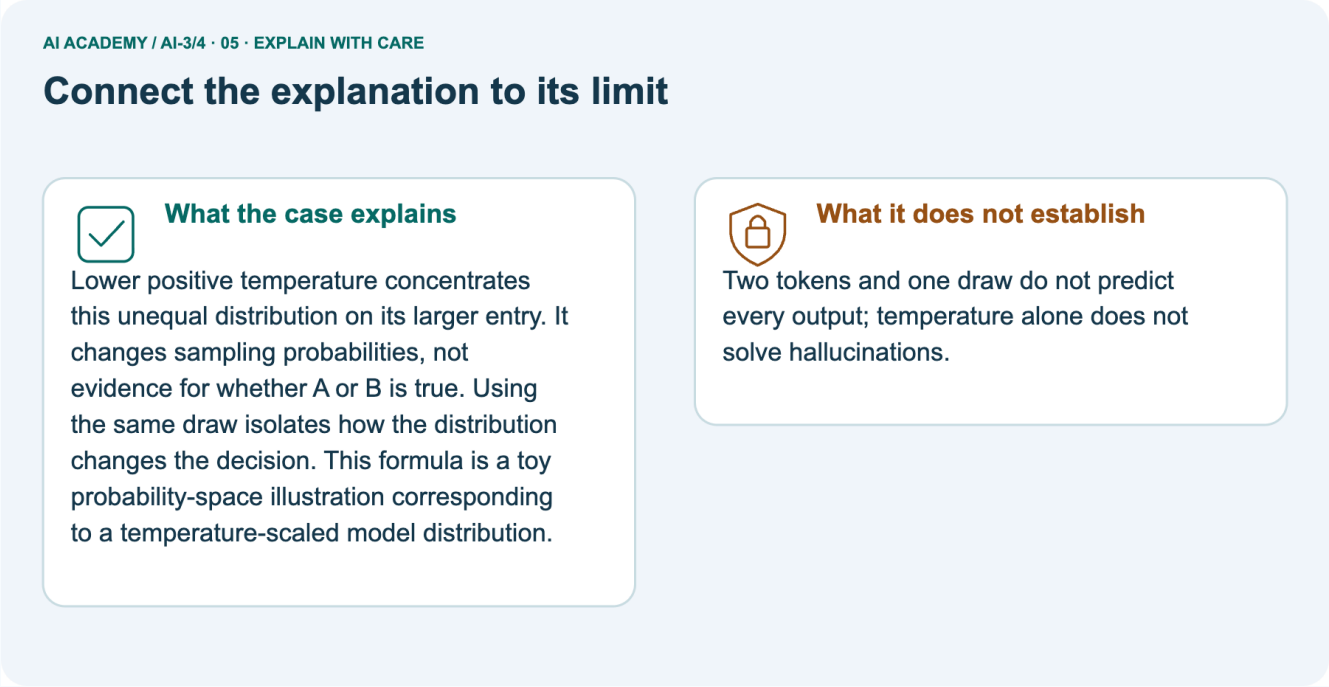
supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Limit: Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

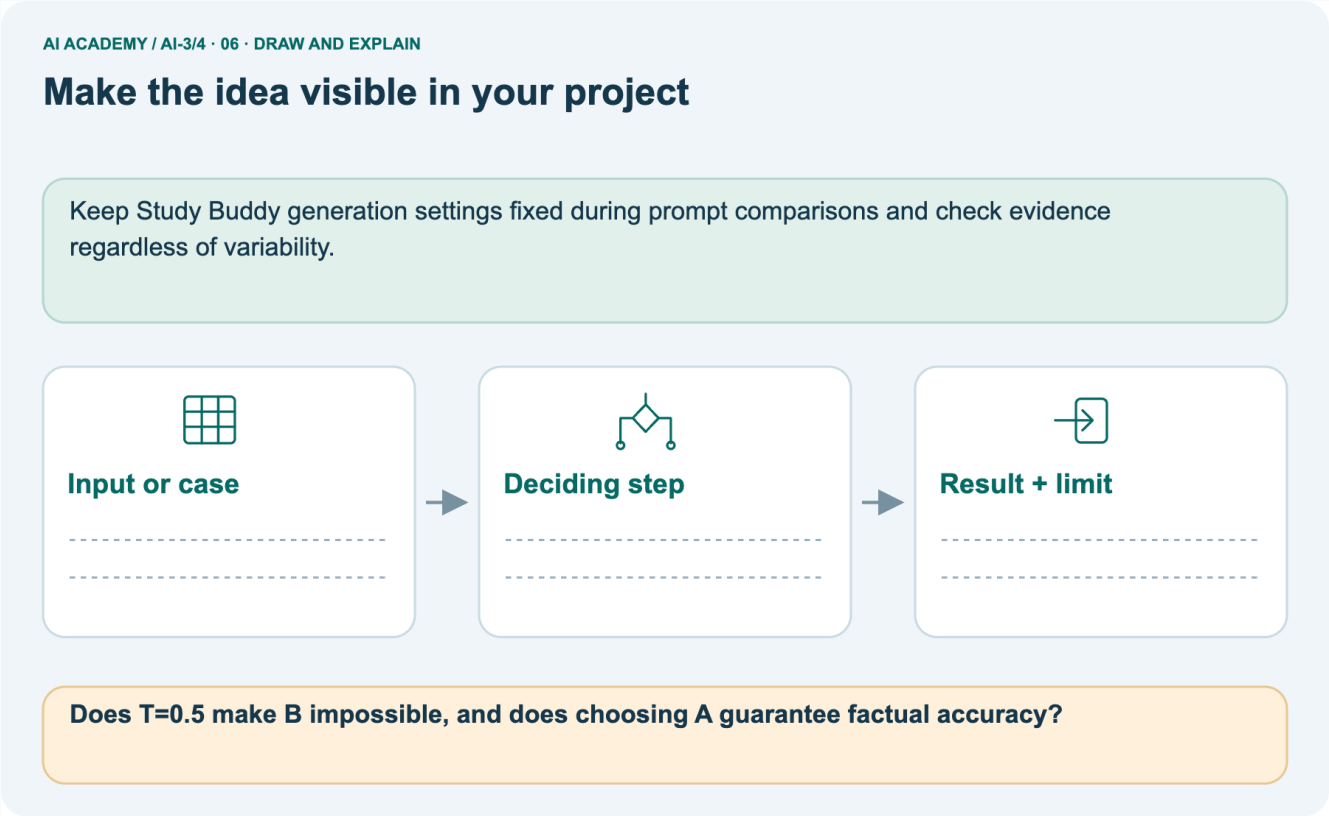
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**. Draw your **input**, deciding step, result and limit.

Supported question: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: No. B still has probability 1/17, and the table supplies no truth labels for either token.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.